

KUBERNETES CON LOAD BALANCER Y AWS



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0.1 - Instalar k3s (Kubernetes ultraligero – SIN Docker).

Actualizar sistema

WSL no me permitía realizar el comando completo, por lo que decido separarlo en dos partes.

```
alumno04@A6Alumno04:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
120 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
alumno04@A6A1umno04:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following package was automatically installed and is no longer required:
  libllvm19

Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  libllvm20

The following packages will be upgraded:
  apparmor binutils binutils-common binutils-x86-64-linux-gnu bsdxextrautils bsdutils cloud-init coreutils
  dconf-gsettings-backend dconf-service dhcpcd-base dirnmgr distro-info-data dpkg eject fdisk gir1.2-glib-2.0
  gir1.2-packagekitglib-1.0 gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpgconf gpgsm gpgv keyboxd
  landscape-client landscape-common libapparmor1 libavahi-client3 libavahi-common-data libavahi-common3 libbinutils
  libblkid1 libc-bin libc6 libc6tcf-nobfd0 libc6tcf0 libcups2t64 libdbusconf1 libdrm-amdgpu1 libdrm-common libdrm-intel1
  libdrm2 libegl-mesa0 libfdisk1 libgbm1 libgl1-mesa-dri libglib2.0-0t64 libglib2.0-bin libglib2.0-data
  libglx-mesa0 libgprofng0 libmount1 libnetplan1 libnss-systemd libpackagekit-glib2-18 libpam-modules
  libpam-modules-bin libpam-runtime libpam-systemd libpam0g libpng16-16t64 libpython3-stdlib libpython3.12-minimal
  libpython3.12-stdlib libpython3.12t64 libsqlite3 libsmartcols1 libsodium23 libsqlite3-0 libssh-4 libssl3t64
  libsystemd-shared libsystemd0 libtasn1-6 libtiff6 libudev1 libuuid1 libxml2 locales mesa-libgallium
  mesa-vulkan-drivers mount netplan-generator netplan.io openssl-client openssl packagekit packagekit-tools
  python-apt-common python3 python3-apt python3-minimal python3-netplan python3-netplan python3-software-properties python3-urllib3
  python3.12 python3.12-minimal snapd software-properties-common systemd systemd-dev systemd-hwe-hwdb
  systemd-resolved systemd-sysv systemd-timesyncd ubuntu-pro-client ubuntu-pro-client-l10n udev util-linux
  uuid-runtime vim vim-common vim-runtime vim-tiny wsl-pro-service xxd

120 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
60 standard LTS security updates
Need to get 155 MB of archives.
After this operation, 149 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Instalar dependencias mínimas

`sudo apt install -y curl wget git`

```
alumno04@A6Alumno04:~$ sudo apt install -y curl wget git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Instalar k3s SIN systemd (mejor para WSL2)

`curl -sL https://get.k3s.io | K3S_KUBECONFIG_MODE="644" sh -`

```
alumno04@A6Alumno04:~$ curl -sL https://get.k3s.io | K3S_KUBECONFIG_MODE="644" sh -
[INFO] Finding release for channel stable
[INFO] Using v1.34.3+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.3+k3s1/sha256sum-amd64.txt
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.34.3+k3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectrl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s.service → /etc/systemd/system/k3s.service.
[INFO] Host iptables-save/iptables-restore tools not found
[INFO] Host ip6tables-save/ip6tables-restore tools not found
[INFO] systemd: Starting k3s
```

Verificar que k3s está instalado

`sudo k3s --version`

Se comprueba la versión de k3s: **versión v1.34.3+k3s1**

```
alumno04@A6Alumno04:~$ sudo k3s --version
k3s version v1.34.3+k3s1 (48ffa7b6)
go version go1.24.11
```

Iniciar k3s y esperar 10 segundos a que inicie

sudo k3s server & sleep 10

```
alumno04@A6Alumno04:~$ sudo k3s server & sleep 10
[1] 8777
INFO[0000] Starting k3s v1.34.3+k3s1 (48ffa7b6)
INFO[0000] Configuring sqlite3 database connection pooling: maxIdleConns=2, maxOpenConns=0, connMaxLifetime=0s
INFO[0000] Configuring database table schema and indexes, this may take a moment...
INFO[0000] Database tables and indexes are up to date
INFO[0000] Kine available at unix:///kine.sock
INFO[0000] Reconciling bootstrap data between datastore and disk
INFO[0000] Password verified locally for node a6alumno04
INFO[0000] certificate CN=a6alumno04 signed by CN=k3s-server-ca@1768915366: notBefore=2026-01-20 13:22:46 +0000 UTC notAfter=2027-01-20 13:24:20 +0000 UTC
INFO[0000] certificate CN=system:kube-proxy signed by CN=k3s-client-ca@1768915366: notBefore=2026-01-20 13:22:46 +0000 UTC notAfter=2027-01-20 13:24:20 +0000 UTC
INFO[0000] Module overlay was already loaded
INFO[0000] Module nf_conntrack was already loaded
INFO[0000] Module br_netfilter was already loaded
INFO[0000] Module iptable_nat was already loaded
INFO[0000] Module iptable_filter was already loaded
WARN[0000] Failed to load kernel module nft-expr-counter with modprobe
INFO[0000] Connecting to proxy url="wss://127.0.0.1:6443/v1-k3s/connect"
INFO[0000] Creating k3s-cert-monitor event broadcaster
INFO[0000] Logging containerd to /var/lib/rancher/k3s/agent/containerd/containerd.log
INFO[0000] Running containerd -c /var/lib/rancher/k3s/agent/etc/containerd/config.toml
INFO[0000] Connection to etcd is ready
INFO[0000] ETCD server is now running
INFO[0000] Running kube-apiserver --advertise-port=6443 --allow-privileged=true --anonymous-auth=false --api-audience
```

Verificar que está corriendo

sudo k3s kubectl get nodes

```
alumno04@A6Alumno04:~$ sudo k3s kubectl get nodes
NAME           STATUS    ROLES    AGE     VERSION
a6alumno04     Ready    control-plane  2m13s   v1.34.3+k3s1
```

0.2 - Instalar Kubectl localmente (para comodidad).

#Descargar kubectl

```
- curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

Se muestra el resultado del comando **curl**:

```
alumno04@A6Alumno04:~$ curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0    0     0      0      0  --:--:--  0:00:00  --:--:--     0curl: (6) Could not resolve host: xn--s
-5gn
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
100  138    100  138    0     0   361      0  --:--:--  --:--:--  --:--:--   361
100    7    100    7    0     0    10      0  --:--:--  --:--:--  --:--:--    10
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
100  138    100  138    0     0   611      0  --:--:--  --:--:--  --:--:--   613
100 55.8M    100 55.8M    0     0 59.7M      0  --:--:--  --:--:--  --:--:--  95.7M
```

```
- sudo chmod +x kubectl
```

```
- sudo mv kubectl /usr/local/bin/
```

```
alumno04@A6Alumno04:~$ sudo chmod +x kubectl
alumno04@A6Alumno04:~$ sudo mv kubectl /usr/local/bin
```

#Verificar

Kubectl version --client

```
alumno04@A6Alumno04:~$ kubectl version --client
Client Version: v1.35.0
Kustomize Version: v5.7.1
```

#Configurar kubeconfig para acceder sin sudo

```
- mkdir -p $HOME/.kube
```

```
- sudo cp /etc/rancher/k3s/k3s.yaml $HOME/.kube/config
```

```
- sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

```
- sudo chmod 600 $HOME/.kube/config
```

Se crea el directorio, se copia el contenido del yaml de k3s y se le dan permisos al **config**:

```
alumno04@A6Alumno04:~$ mkdir -p $HOME/.kube
alumno04@A6Alumno04:~$ sudo cp /etc/rancher/k3s/k3s.yaml $HOME/.kube/config
alumno04@A6Alumno04:~$ sudo chown $(id -u):$(id -g) $HOME/.kube/config
alumno04@A6Alumno04:~$ sudo chmod 600 $HOME/.kube/config
```

#Verificar que funciona sin ruido

Kubectl get nodes

```
alumno04@A6Alumno04:~$ kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
a6alumno04    Ready     control-plane  18m   v1.34.3+k3s1
```

0.3. Crear directorio de trabajo.

- mkdir -p ~/kubernetes-aws-practice

- cd ~/kubernetes-aws-practice

Se crea el directorio y se accede al mismo:

```
alumno04@A6Alumno04:~$ mkdir -p ~/kubernetes-aws-practice
alumno04@A6Alumno04:~$ cd ~/kubernetes-aws-practice/
alumno04@A6Alumno04:~/kubernetes-aws-practice$
```

PARTE 1. CREAR APLICACIÓN SIMPLE PARA KUBERNETES.

1.1. Crear carpeta para la aplicación.

```
- mkdir -p ~/kubernetes-aws-practica/app  
- cd ~/kubernetes-aws-practice/app
```

Al crear el directorio no saqué captura del resultado, pero muestro captura del historial de la terminal:

```
42 mkdir -p ~/kubernetes-aws-practice/app  
43 cd ~/kubernetes-aws-practice/app/
```

1.2. Crear aplicación para Python con Flask.

Crear app.py:

```
GNU nano 7.2 app.py  
#!/usr/bin/env python3  
  
import os  
import sys  
import socket  
from datetime import datetime  
from flask import Flask, jsonify, send_from_directory  
  
# Inicialización de la app  
app = Flask(__name__)  
  
# Variables de entorno inyectadas por Kubernetes  
POD_NAME = os.getenv("POD_NAME", "Unknown Pod")  
POD_NAMESPACE = os.getenv("POD_NAMESPACE", "default")  
  
@app.route("/")  
def index():  
    """Sirve el archivo index.html"""  
    return send_from_directory(".", "index.html")  
  
@app.route("/pod-info")  
def pod_info():  
    """Devuelve información del pod"""  
    return jsonify({  
        "pod_name": POD_NAME,  
        "namespace": POD_NAMESPACE,  
        "hostname": socket.gethostname(),  
        "timestamp": datetime.now().isoformat()  
    })  
  
@app.route("/health")  
def health():  
    """Endpoint de salud"""  
    return jsonify({  
        "status": "healthy",  
        "pod": POD_NAME  
    }), 200  
  
if __name__ == "__main__":  
    print(f"[{POD_NAME}] Iniciando servidor Flask...", file=sys.stderr)  
    app.run(host="0.0.0.0", port=5000, debug=False)
```

Se asignan permisos de ejecución:

```
alumno04@A6Alumno04:~/kubernetes-aws-practice/app$ sudo chmod +x app.py
```

1.3. Crear archivo HTML.

Se muestra parte del contenido del archivo HTML para demostrar su correcta creación:

```
GNU nano 7.2 index.html *
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Kubernetes Load Balancer</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    }

    .container {
      background: white;
      padding: 50px;
      border-radius: 10px;
      box-shadow: 0 10px 25px rgba(0, 0, 0, 0.2);
      text-align: center;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>Kubernetes Load Balancer</h1>
  </div>
</body>
</html>
```

1.4 - Crear requirements.txt.

- cat > requirements.txt << 'EOF'
- flask==3.0.0
- werkzeug==3.0.0

Se realiza mediante nano porque con **cat** no se configura correctamente:

```
GNU nano 7.2 requirements.txt
Flask==3.0.0
Werkzeug==3.0.0
```

1.5 - Crear dockerfile ultraligero.

Nota, este Dockerfile es solo para construcción local. K3s lo ejecutará directamente:
Se realiza mediante **sudo nano** porque con cat sigue dando error:

```
GNU nano 7.2                                dockerfile
# Imagen base
FROM python:3.11-slim

# Establecer directorio de trabajo
WORKDIR /app

# Copiar archivos de la aplicación
COPY requirements.txt .
COPY app.py .
COPY index.html .

# Instalar dependencias de Python
RUN pip install --no-cache-dir -r requirements.txt

# Comando por defecto para ejecutar la aplicación
CMD ["python", "app.py"]
```

PARTE 2. CONFIGURAR KUBERNETES (MANIFESTS YAML)

2.1 - Crear Namespace.

cd ~/kubernetes-aws-practice

Se reestructura el código para evitar errores y se crea directamente el fichero mediante **sudo nano namespace.yaml**:

```
GNU nano 7.2                                namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: load-balancer-demo
  labels:
    name: load-balancer-demo
```

#Aplicar

- kubectl apply -f namespace

#Verificar

- kubectl get namespaces

Se muestra el resultado de los comandos:

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl apply -f namespace.yaml
namespace/load-balancer-demo created
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl get namespaces
NAME                STATUS    AGE
default             Active    47h
kube-node-lease     Active    47h
kube-public          Active    47h
kube-system          Active    47h
load-balancer-demo  Active    8s
alumno04@A6Alumno04:~/kubernetes-aws-practice$
```

2.2 - Crear ConfigMap con archivos de la app.

Se muestra un fragmento del archivo que contiene todo el código:

```
GNU nano 7.2                                configmap.yaml *
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-files
  namespace: load-balancer-demo
data:
  requirements.txt: |
    Flask==3.0.0
    Werkzeug==3.0.0

  app.py: |
    #!/usr/bin/env python3
    from flask import Flask, jsonify, send_from_directory
    import os
    import socket
    from datetime import datetime
    import sys

    app = Flask(__name__)

    POD_NAME = os.getenv('POD_NAME', 'Unknown Pod')
    POD_NAMESPACE = os.getenv('POD_NAMESPACE', 'default')

    @app.route('/')
    def index():
        return send_from_directory('.', 'index.html')

    @app.route('/pod-info')
    def pod_info():
        return jsonify({
            'pod_name': POD_NAME,
            'namespace': POD_NAMESPACE,
            'hostname': socket.gethostname(),
            'timestamp': datetime.now().isoformat()
        })

    @app.route('/health')
    def health():
        return jsonify({'status': 'healthy', 'pod': POD_NAME}), 200

    if __name__ == '__main__':
        print(f"[{POD_NAME}] Iniciando servidor Flask...", file=sys.stderr)
        app.run(host='0.0.0.0', port=5000, debug=False)

  index.html: |
    <!DOCTYPE html>
    <html lang="es">
    <head>
      <meta charset="UTF-8">
      <title>Kubernetes Load Balancer</title>
      <style>
        body {
          font-family: Arial, sans-serif;
          display: flex;
          justify-content: center;
          align-items: center;
          min-height: 100vh;
          margin: 0;
        }
      </style>
    </head>
    <body>
      <div>
        <h1>Kubernetes Load Balancer</h1>
        <p>Este es un servidor Flask que se ejecuta en un pod de Kubernetes.
        </p>
      </div>
    </body>
  </html>

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo
```

2.3 - Crear deployment con 3 réplicas.

Se muestra parte de la configuración del fichero deployment.yaml:

```
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-app
  namespace: load-balancer-demo
  labels:
    app: web-app
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-app
  template:
    metadata:
      labels:
        app: web-app
    spec:
      containers:
        - name: web-app
          image: python:3.11-slim
          command: ["sh", "-c"]
          args:
            - |
              cd /app
              pip install --no-cache-dir -r requirements.txt > /dev/null 2>&1
              python app.py
          ports:
            - containerPort: 5000
              protocol: TCP
          env:
            - name: POD_NAME
              valueFrom:
                fieldRef:
                  fieldPath: metadata.name
            - name: POD_NAMESPACE
              valueFrom:
                fieldRef:
                  fieldPath: metadata.namespace
          volumeMounts:
            - name: app-volume
              mountPath: /app
      livenessProbe:
        httpGet:
          path: /health
          port: 5000
          initialDelaySeconds: 15
          periodSeconds: 10
      readinessProbe:
        httpGet:
          path: /health
          port: 5000
          initialDelaySeconds: 5
          periodSeconds: 5
      volumes:
        - name: app-volume
          configMap:
            name: app-files
            defaultMode: 0755
```

Aplicar deployment

Kubectl apply -f deployment.yaml

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl apply -f deployment.yaml
deployment.apps/web-app created
```

Verificar que los pods se están creando

Kubectl get pods -n load-balancer-demo

```

alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl get pods -n load-balancer-demo
NAME                                READY   STATUS    RESTARTS   AGE
web-app-65967466dd-d96pz           1/1     Running   0           19h
web-app-65967466dd-lxqcd           1/1     Running   0           19h
web-app-65967466dd-z2bgp           1/1     Running   0           19h

```

#esperar a que estén ready

- echo "Esperando a que los pods estén listos..."

- kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo --timeout=120s

```

alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo --timeout=120s
timed out waiting for the condition on pods/web-app-65967466dd-d96pz
timed out waiting for the condition on pods/web-app-65967466dd-lxqcd
timed out waiting for the condition on pods/web-app-65967466dd-z2bgp

```

Verificar.

Kubectl get pods --load-balancer-demo

```

alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl get pods -n load-balancer-demo
NAME                                READY   STATUS    RESTARTS   AGE
web-app-65967466dd-d96pz           1/1     Running   1 (15m ago)  20h
web-app-65967466dd-lxqcd           1/1     Running   1 (15m ago)  20h
web-app-65967466dd-z2bgp           1/1     Running   1 (15m ago)  20h

```

2.4 - Crear service (Load Balancer)

```

GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: web-app-service
  namespace: load-balancer-demo
  labels:
    app: web-app
spec:
  type: LoadBalancer
  selector:
    app: web-app
  ports:
    - name: http
      protocol: TCP
      port: 80
      targetPort: 5000
  sessionAffinity: None

```

#Aplicar service

Kubectl apply -f service.yaml

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl apply -f service.yaml
service/web-app-service created
```

#Verificar el service

Kubectl get svc -n load-balancer-demo

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl get svc -n load-balancer-demo
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
web-app-service     LoadBalancer  10.43.94.143  <pending>    80:31076/TCP     28s
```

#Obtener IP del service

Kubectl get svc -n load-balancer-demo web-app-service -o wide

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl get svc -n load-balancer-demo web-app-service -o wide
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE  SELECTOR
web-app-service     LoadBalancer  10.43.94.143  <pending>    80:31076/TCP     58s  app=web-app
```

PARTE 3. VERIFICAR BALANCEO DE CARGA LOCAL.

3.1 - Levantar la aplicación.

Opción 1:Port-forward (recomendado para WSL2)

kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80

Se introduce el siguiente comando y se muestra el resultado:

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80
Forwarding from 127.0.0.1:8080 -> 5000
Forwarding from [::1]:8080 -> 5000
```

3.2. Probar balanceo con curl.

Script para hacer peticiones y ver qué pod responde

```
GNU nano 7.2 balanceo.sh
for i in {1..10}; do
  echo "Petición $i:"
  curl -s http://localhost:8080/pod-info | python3 -m json.tool | grep pod_name
  sleep 1
done
```

Se le dan permisos de ejecución al script y se ejecuta:

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ ./balanceo.sh
Petición 1:
  "pod_name": "web-app-65967466dd-d96pz",
Petición 2:
  "pod_name": "web-app-65967466dd-z2bgp",
Petición 3:
  "pod_name": "web-app-65967466dd-z2bgp",
Petición 4:
  "pod_name": "web-app-65967466dd-d96pz",
Petición 5:
  "pod_name": "web-app-65967466dd-z2bgp",
Petición 6:
  "pod_name": "web-app-65967466dd-z2bgp",
Petición 7:
  "pod_name": "web-app-65967466dd-d96pz",
Petición 8:
  "pod_name": "web-app-65967466dd-d96pz",
Petición 9:
  "pod_name": "web-app-65967466dd-z2bgp",
Petición 10:
  "pod_name": "web-app-65967466dd-d96pz",
```

En la imagen previa se muestra cómo cambia el pod que atiende la solicitud, ya que cambia su terminación.

PARTE 4. CONFIGURACIÓN EN AWS.

4.1. Crear Security Group.

En AWS Console:

1. **Accede a EC2:** Security Groups.
2. Haz clic en “Create security group”
3. **Nombre:** kubernetes-aws-sg
4. **Descripción:** Security group para Kubernetes con AWS.

Agregar reglas de entrada:

Tipo	Protocolo	Puerto	Origen
SSH	TCP	22	0.0.0.0/0
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

Nombre y descripción:

Detalles básicos

Nombre del grupo de seguridad [Información](#)

El nombre no se puede editar después de su creación.

Descripción [Información](#)

VPC [Información](#)

Reglas de entrada:

Reglas de entrada [Información](#)

Tipo Información	Protocolo Información	Intervalo de puertos Información	Origen Información	Descripción: opcional Información
SSH	TCP	22	A... 0.0.0.0/0	 Eliminar
HTTP	TCP	80	A... 0.0.0.0/0	 Eliminar
HTTPS	TCP	443	A... 0.0.0.0/0	 Eliminar

4.2 – Crear instancia EC2

En AWS Console:

1. **EC2** → Instances → Launch instances
2. **Nombre:** kubernetes-test-server

Nombre y etiquetas [Información](#)

Nombre
 [Agregar etiquetas adicionales](#)

3. AMI: Ubuntu 24.04 LTS

▼ Imágenes de aplicaciones y sistemas operativos (Imagen de máquina de Amazon) [Información](#)

Una AMI posee el sistema operativo, el servidor de aplicaciones y las aplicaciones de la instancia. Si a continuación no ve una AMI adecuada, utilice el campo de búsqueda o elija **Buscar más AMI**.

Recientes

Inicio rápido

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

...

>

🔍

Buscar más AMI

Inclusión de AMI de AWS, Marketplace y la comunidad

Imágenes de máquina de Amazon (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0b6c6ebd2801a5cb (64 bits (x86)) / ami-0071c8c431eea0edb (64 bits (Arm))
Virtualización: hvm Activado para ENA: true Tipo de dispositivo raíz: ebs

Apto para la capa gratuita

Descripción

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Arquitectura	ID de AMI	Fecha de publicación	Nombre de usuario	
64 bi... ▼	ami-0b6c6ebd2801a5cb	2025-12-12	ubuntu	Proveedor verificado

4. Tipo: t3.micro (Free Tier)

▼ Tipo de instancia [Información](#) | [Obtener asesoramiento](#)

Tipo de instancia

t3.micro Apto para la capa gratuita

Familia: t3 2 vCPU 1 GiB Memoria Generación actual: true

Bajo demanda Ubuntu Pro base precios: 0.0139 USD por hora

Bajo demanda SUSE base precios: 0.0104 USD por hora

Bajo demanda Linux base precios: 0.0104 USD por hora

Bajo demanda RHEL base precios: 0.0392 USD por hora

Bajo demanda Windows base precios: 0.0196 USD por hora

☐ Todas las generaciones

[Comparar tipos de instancias](#)

Se aplican costos adicionales a las AMI con software preinstalado

5. Key pair: Tu clave SSH

▼ Par de claves (inicio de sesión) [Información](#)

Puede utilizar un par de claves para conectarse de forma segura a la instancia. Asegúrese de que tiene acceso al par de claves seleccionado antes de lanzar la instancia.

Nombre del par de claves - *obligatorio*

vockey

▼

↻

[Crear un nuevo par de claves](#)

6. VPC: Default

7. Security group: kubernetes-aws-sg

▼ Configuraciones de red [Información](#) [Editar](#)

Red [Información](#)

vpc-0f583dcf81a9ded72

Subred [Información](#)

Sin preferencias (subred predeterminada en cualquier zona de disponibilidad)

Asignar automáticamente la IP pública [Información](#)

Habilitar

Firewall (grupos de seguridad) [Información](#)

Un grupo de seguridad es un conjunto de reglas de firewall que controlan el tráfico de la instancia. Agregue reglas para permitir que un tráfico específico llegue a la instancia.

☐ Crear grupo de seguridad

☒ Seleccionar un grupo de seguridad existente

Grupos de seguridad comunes [Información](#)

Seleccionar grupos de seguridad

▼

kubernetes-aws-sg sg-017cfc69283e4cc21 ✕

VPC: vpc-0f583dcf81a9ded72

↻

[Compare reglas de grupo de seguridad](#)

Los grupos de seguridad que agrega o elimine aquí se agregarán a todas las interfaces de red o se eliminarán de ellas.

8. Storage: 8 GiB, gp3

▼ Configurar almacenamiento

Información

Avanzado

1x GiB Volumen raíz, 3000 IOPS, No cifrado

Agregar un nuevo volumen

La AMI seleccionada contiene volúmenes de almacén de instancias; sin embargo, la instancia no permite dichos volúmenes. Por lo que no se podrá obtener acceso a ninguno de estos volúmenes de la AMI desde la instancia

Haga clic en actualizar para ver la información de la copia de seguridad

Haga clic en actualizar para ver la información de la copia de seguridad

Las etiquetas que asigne determinan si alguna política de Data Lifecycle Manager realizará una copia de seguridad de la instancia.

0 x sistemas de archivos

Editar

9. Launch instance.

▼ Resumen

Número de instancias

Información

Imagen de software (AMI)

Canonical, Ubuntu, 24.04, amd64...más información

ami-0b6c6ebed2801a5cb

Tipo de servidor virtual (tipo de instancia)

t3.micro

Firewall (grupo de seguridad)

kubernetes-aws-sg

Almacenamiento (volúmenes)

Volúmenes: 1 (8 GiB)

Cancelar

Lanzar instancia

Código de versión preliminar

4.3 – Conectar a EC2

Obtener IP pública desde AWS Console (ej: 54.123.45.67)

Conectar via SSH `ssh -i ~/.ssh/tu-clave-aws.pem ubuntu@54.123.45.67`

```
alumno04@A6Alumno04:~/.ssh$ ssh -i ~/.ssh/labsuser.pem ubuntu@44.193.84.245
The authenticity of host '44.193.84.245 (44.193.84.245)' can't be established.
ED25519 key fingerprint is SHA256:mSGXq3D0qBALixCF3TRnP4gqNXxxIzSL+2JAg4xPZ8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.193.84.245' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jan 27 08:58:58 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.0% of 6.71GB Processes:      109
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.9.35

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
```

En EC2, instalar herramientas

`sudo apt update`

Se actualizan los paquetes de la instancia:

```
ubuntu@ip-172-31-9-35:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1410 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [229 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9820 B]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [924 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [209 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2302 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [527 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.0 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6472 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
```

```
ubuntu@ip-172-31-9-35:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  libnuma1 numactl
The following packages will be upgraded:
  bsdxtrautils bsduutils dirmngr eject fdisk gir1.2-glib-2.0 gnupg gnupg-l10n gnupg-utils gpg gpg-agent
  gpg-wks-client gpgconf gpgsm gpgv keyboxd klibc-utils kpartx libblkid1 libfdisk1 libglib2.0-0t64
  libglib2.0-bin libglib2.0-data libklibc libmbim-glib4 libmbim-proxy libmbim-utils libmount1
  libnss-systemd libpam-systemd libpng16-16t64 libpython3.12-minimal libpython3.12-stdlib libpython3.12t64
  libsmartcols1 libsodium23 libsystemd-shared libsystemd0 libtasn1-6 libudev1 libuuid1 libxml2 libxslt1.1
  mount multipath-tools python3-pyasn1 python3-urllib3 python3.12 python3.12-minimal screen snapd systemd
  systemd-dev systemd-resolved systemd-sysv udev util-linux uuid-runtime
58 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
30 standard LTS security updates
Need to get 60.8 MB of archives.
After this operation, 649 kB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

sudo apt install -y curl wget python3 python3-pip git

Se instalan las herramientas necesarias:

```
ubuntu@ip-172-31-9-35:~$ sudo apt install -u curl wget python3 python3-pip git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp cpp-13
  cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev fakeroot fontconfig-config fonts-dejavu-core
  fonts-dejavu-mono g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu javascript-common libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3 libasan8 libatomic1 libbinutils libc-dev-bin
  libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl
  libexpat1-dev libfakeroot libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0
  libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhwasan0 libisl23 libitm1
  libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3
  libpython3-dev libpython3.12-dev libquadmath0 libsqlite3-dev libstdc++-13-dev libtiff6 libtsan2
  libubsan1 libwebp7 libxpm4 linux-libc-dev lto-disabled-list make manpages-dev python3-dev python3-wheel
  python3.12-dev rpcsvc-proto zlib1g-dev
Suggested packages:
  binutils-doc gprofng-gui bzip2-doc cpp-doc gcc-13-locales cpp-13-doc debian-keyring g++-multilib
  g++-13-multilib gcc-13-doc gcc-multilib autoconf automake libtool flex bison gdb gcc-doc gcc-13-multilib
  gdb-x86-64-linux-gnu apache2 | lighttpd | httpd glibc-doc bzip2 libgd-tools libheif-plugin-x265
  libheif-plugin-ffmpegdec libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec
  libheif-plugin-j2kenc libheif-plugin-rav1e libheif-plugin-svtenc libstdc++-13-doc make-doc
```

Crear directorio de trabajo

mkdir -p ~/kubernetes-test

Se crea el directorio kubernetes-test:

```
ubuntu@ip-172-31-9-35:~$ mkdir -p ~/kubernetes-test
ubuntu@ip-172-31-9-35:~$
```

PARTE 5. CONECTAR KUBERNETES LOCAL CON AWS EC2

5.1 - Crear túnel SSH que expone el LoadBalancer

En una nueva máquina local (WSL2), mantén esta terminal abierta, el túnel estará activo mientras esté abierta:

```
alumno04@A6Alumno04:~/kubernetes-aws-practice$ sudo ssh -i ~/.ssh/labsuser.pem -N -R 8888:localhost:8080 ubuntu@3.218.143.63
[sudo] password for alumno04:
The authenticity of host '3.218.143.63 (3.218.143.63)' can't be established.
ED25519 key fingerprint is SHA256:mSGXq3D0qBALixCF3TRnP4gqNXxxIzSL+2JAKg4xPZ8.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.218.143.63' (ED25519) to the list of known hosts.
```

En AWS EC2 (otra terminal local):

#Conecta a la instancia en otra terminal

```
- ssh -i ~/.ssh/tu-clave-aws.pem ubuntu@54.123.45.67
```

#Verificar que el túnel funciona

```
- curl -s http://localhost:8888/pod-info
```

#Deberías recibir json:

```
- {"pod_name":"web-app-xxxxx-11111","namespace":"load-balancer-demo",...}
```

Se ejecuta curl y devuelve el resultado esperado:

```
ubuntu@ip-172-31-9-35:~$ curl -s http://localhost:8888/pod-info
{"hostname":"web-app-65967466dd-lxqcd","namespace":"load-balancer-demo","pod_name":"web-app-65967466dd-lxqcd","timestamp":"2026-01-29T13:36:21.200430"}
```

5.2 - Crear script de prueba en EC2.

En la instancia EC2:

Se crea el script test-kubernetes-lb.sh:

```
GNU nano 7.2 /home/ubuntu/test-kubernetes-lb.sh
#!/bin/bash

echo "=== Prueba Kubernetes Load Balancer desde AWS ==="
echo ""

declare -A pods_count

for i in {1..15}; do
    # Llamada al endpoint del Load Balancer
    response=$(curl -s http://localhost:8888/pod-info)

    # Extraer el nombre del pod usando Python
    pod_name=$(echo "$response" | python3 -c "
import sys, json
try:
    data = json.load(sys.stdin)
    print(data.get('pod_name', 'ERROR'))
except:
    print('ERROR')
")

    echo "Petición $i → Pod: $pod_name"

    # Contar cuántas veces responde cada pod
    pods_count[$pod_name]=$(( ${pods_count[$pod_name]:-0} + 1 ))

    sleep 1
done

echo ""
echo "=== Resumen Balanceo ==="
for pod in "${!pods_count[@]}; do
    echo "Pod $pod: ${pods_count[$pod]} peticiones"
done
```

Se dan permisos de ejecución al script y se ejecuta, devolviendo el nombre del pod después de realizar la petición.

```
ubuntu@ip-172-31-9-35:~$ sudo chmod +x ~/test-kubernetes-lb.sh
ubuntu@ip-172-31-9-35:~$ ~/test-kubernetes-lb.sh
=== Prueba Kubernetes Load Balancer desde AWS ===

Petición 1 → Pod: web-app-65967466dd-lxqcd
Petición 2 → Pod: web-app-65967466dd-lxqcd
Petición 3 → Pod: web-app-65967466dd-lxqcd
Petición 4 → Pod: web-app-65967466dd-lxqcd
Petición 5 → Pod: web-app-65967466dd-lxqcd
Petición 6 → Pod: web-app-65967466dd-lxqcd
Petición 7 → Pod: web-app-65967466dd-lxqcd
Petición 8 → Pod: web-app-65967466dd-lxqcd
Petición 9 → Pod: web-app-65967466dd-lxqcd
Petición 10 → Pod: web-app-65967466dd-lxqcd
Petición 11 → Pod: web-app-65967466dd-lxqcd
Petición 12 → Pod: web-app-65967466dd-lxqcd
Petición 13 → Pod: web-app-65967466dd-lxqcd
Petición 14 → Pod: web-app-65967466dd-lxqcd
Petición 15 → Pod: web-app-65967466dd-lxqcd

=== Resumen Balanceo ===
Pod web-app-65967466dd-lxqcd: 15 peticiones
```

PARTE 6. ESCALADO DINÁMICO

6.1 - Escalar a 5 réplicas.

En tu máquina local:

Aumentar a 5 réplicas.

Kubectrl scale deployment web-app -n load-balancer-demo --replicas=5

```
alumno04@A6Alumno04:~$ kubectl scale deployment web-app -n load-balancer-demo --replicas=5
deployment.apps/web-app scaled
```

#Verificar

Kubectrl get pods -n load-balancer-demo

Se muestra como los pods están “running”

```
alumno04@A6Alumno04:~$ kubectl get pods -n load-balancer-demo
NAME                                READY   STATUS    RESTARTS   AGE
web-app-65967466dd-4vpxm           1/1     Running   0           15s
web-app-65967466dd-d96pz           1/1     Running   4 (5h11m ago)  7d
web-app-65967466dd-f9jqm           1/1     Running   0           15s
web-app-65967466dd-lxqcd           1/1     Running   4 (5h11m ago)  7d
web-app-65967466dd-z2bgp           1/1     Running   4 (5h11m ago)  7d
alumno04@A6Alumno04:~$
```

Esperar a que estén ready

Kubectrl wait --for=condition=ready pod -l app=web -n load-balancer-demo --timeout=120s

Devuelve como resultado que la condición se cumple:

```
alumno04@A6Alumno04:~$ kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo --timeout=120s
pod/web-app-65967466dd-4vpxm condition met
pod/web-app-65967466dd-d96pz condition met
pod/web-app-65967466dd-f9jqm condition met
pod/web-app-65967466dd-lxqcd condition met
pod/web-app-65967466dd-z2bgp condition met
```

6.2 - Probar balanceo desde AWS

En EC2:

```
~/test-kubernetes-lb.sh
```

Devuelve como resultado de las peticiones el nombre del pod. Esto indica que hay comunicación y respuesta entre pod y cliente:

```
ubuntu@ip-172-31-9-35:~$ ~/test-kubernetes-lb.sh
=== Prueba Kubernetes Load Balancer desde AWS ===

Petición 1 → Pod: web-app-65967466dd-lxqcd
Petición 2 → Pod: web-app-65967466dd-lxqcd
Petición 3 → Pod: web-app-65967466dd-lxqcd
Petición 4 → Pod: web-app-65967466dd-lxqcd
Petición 5 → Pod: web-app-65967466dd-lxqcd
Petición 6 → Pod: web-app-65967466dd-lxqcd
Petición 7 → Pod: web-app-65967466dd-lxqcd
Petición 8 → Pod: web-app-65967466dd-lxqcd
Petición 9 → Pod: web-app-65967466dd-lxqcd
Petición 10 → Pod: web-app-65967466dd-lxqcd
Petición 11 → Pod: web-app-65967466dd-lxqcd
Petición 12 → Pod: web-app-65967466dd-lxqcd
Petición 13 → Pod: web-app-65967466dd-lxqcd
Petición 14 → Pod: web-app-65967466dd-lxqcd
Petición 15 → Pod: web-app-65967466dd-lxqcd

=== Resumen Balanceo ===
Pod web-app-65967466dd-lxqcd: 15 peticiones
```

PARTE 7. MONITOREO Y OBSERVABILIDAD

7.1. Ver estado de pods.

Ver todos los pods

```
Kubectrl get pods -n load-balancer-demo
```

```
alumno04@A6Alumno04:~$ kubectl get pods -n load-balancer-demo
NAME                                READY   STATUS    RESTARTS   AGE
web-app-65967466dd-4vpxm           1/1     Running   0           3m37s
web-app-65967466dd-d96pz           1/1     Running   4 (5h14m ago)  7d
web-app-65967466dd-f9jqm           1/1     Running   0           3m37s
web-app-65967466dd-lxqcd           1/1     Running   4 (5h14m ago)  7d
web-app-65967466dd-z2bqp           1/1     Running   4 (5h14m ago)  7d
```

Ver logs de un pod específico.

kubectl logs -n load-balancer-demo -l app=web-app -f

Se muestra el resultado de los logs del pod **4vpxm**:

```
alumno04@A6Alumno04:~$ kubectl logs -n load-balancer-demo web-app-65967466dd-4vpxm
[web-app-65967466dd-4vpxm] Iniciando servidor Flask...
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use
*
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.42.0.69:5000
Press CTRL+C to quit
10.42.0.1 - - [30/Jan/2026 11:10:37] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:42] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:47] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:49] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:52] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:57] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:10:59] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:02] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:07] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:09] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:12] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:17] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:19] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:22] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:27] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:29] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:32] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:37] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:39] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:42] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:47] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:49] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:52] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:57] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:11:59] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:02] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:07] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:09] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:12] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:17] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:19] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:22] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:27] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:29] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:32] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:37] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:39] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:42] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:47] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:49] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:52] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:57] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:12:59] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:13:02] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:13:07] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:13:09] "GET /health HTTP/1.1" 200 -
10.42.0.1 - - [30/Jan/2026 11:13:12] "GET /health HTTP/1.1" 200 -
```

```
Kubectl logs -n load-balancer-demo -l app=web-app -f
```

[illegible]

7.2 - Ver estadísticas.

#CPU y memoria de pods

Kubectl top pods -n load-balancer-demo

Se muestra en cada fila un pod (NAME) y su correspondiente memoria (MEMORY).

```
alumno04@A6Alumno04:~$ kubectl top pods -n load-balancer-demo
NAME                                CPU(cores)   MEMORY(bytes)
web-app-65967466dd-4vpxm           1m           25Mi
web-app-65967466dd-d96pz           1m           25Mi
web-app-65967466dd-f9jqm           1m           25Mi
web-app-65967466dd-lxqcd           2m           32Mi
web-app-65967466dd-z2bgp           2m           25Mi
```

#Nodos del clúster

Kubectl top nodes

```
alumno04@A6Alumno04:~$ kubectl top nodes
NAME          CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
a6alumno04    159m        0%       916Mi           11%
```

#Eventos del clúster

Kubectl get events -n load-balancer-demo

```
alumno04@A6Alumno04:~$ kubectl get events -n load-balancer-demo
LAST SEEN   TYPE      REASON      OBJECT                                MESSAGE
21h         Normal    Scheduled    pod/web-app-65967466dd-4vpxm        Successfully assigned load-balancer-demo/
web-app-65967466dd-4vpxm to a6alumno04
21h         Normal    Pulled       pod/web-app-65967466dd-4vpxm        Container image "python:3.11-slim" already present on machine
21h         Normal    Created      pod/web-app-65967466dd-4vpxm        Created container: web-app
21h         Normal    Started      pod/web-app-65967466dd-4vpxm        Started container web-app
29m         Normal    SandboxChanged pod/web-app-65967466dd-4vpxm        Pod sandbox changed, it will be killed and re-created.
29m         Normal    Pulled       pod/web-app-65967466dd-4vpxm        Container image "python:3.11-slim" already present on machine
29m         Normal    Created      pod/web-app-65967466dd-4vpxm        Created container: web-app
29m         Normal    Started      pod/web-app-65967466dd-4vpxm        Started container web-app
27h         Normal    SandboxChanged pod/web-app-65967466dd-d96pz        Pod sandbox changed, it will be killed and re-created.
27h         Normal    Pulled       pod/web-app-65967466dd-d96pz        Container image "python:3.11-slim" already present on machine
27h         Normal    Created      pod/web-app-65967466dd-d96pz        Created container: web-app
27h         Normal    Started      pod/web-app-65967466dd-d96pz        Started container web-app
22h         Normal    SandboxChanged pod/web-app-65967466dd-d96pz        Pod sandbox changed, it will be killed and re-created.
22h         Normal    Pulled       pod/web-app-65967466dd-d96pz        Container image "python:3.11-slim" already present on machine
22h         Normal    Created      pod/web-app-65967466dd-d96pz        Created container: web-app
22h         Normal    Started      pod/web-app-65967466dd-d96pz        Started container web-app
29m         Normal    SandboxChanged pod/web-app-65967466dd-d96pz        Pod sandbox changed, it will be killed and re-created.
29m         Normal    Pulled       pod/web-app-65967466dd-d96pz        Container image "python:3.11-slim" already present on machine
29m         Normal    Created      pod/web-app-65967466dd-d96pz        Created container: web-app
29m         Normal    Started      pod/web-app-65967466dd-d96pz        Started container web-app
21h         Normal    Scheduled    pod/web-app-65967466dd-f9jqm        Successfully assigned load-balancer-demo/
web-app-65967466dd-f9jqm to a6alumno04
21h         Normal    Pulled       pod/web-app-65967466dd-f9jqm        Container image "python:3.11-slim" already present on machine
21h         Normal    Created      pod/web-app-65967466dd-f9jqm        Created container: web-app
21h         Normal    Started      pod/web-app-65967466dd-f9jqm        Started container web-app
29m         Normal    SandboxChanged pod/web-app-65967466dd-f9jqm        Pod sandbox changed, it will be killed and re-created.
29m         Normal    Pulled       pod/web-app-65967466dd-f9jqm        Container image "python:3.11-slim" already present on machine
29m         Normal    Created      pod/web-app-65967466dd-f9jqm        Created container: web-app
29m         Normal    Started      pod/web-app-65967466dd-f9jqm        Started container web-app
27h         Normal    SandboxChanged pod/web-app-65967466dd-lxqcd        Pod sandbox changed, it will be killed and re-created.
27h         Normal    Pulled       pod/web-app-65967466dd-lxqcd        Container image "python:3.11-slim" already present on machine
27h         Normal    Created      pod/web-app-65967466dd-lxqcd        Created container: web-app
27h         Normal    Started      pod/web-app-65967466dd-lxqcd        Started container web-app
22h         Normal    SandboxChanged pod/web-app-65967466dd-lxqcd        Pod sandbox changed, it will be killed and re-created.
22h         Normal    Pulled       pod/web-app-65967466dd-lxqcd        Container image "python:3.11-slim" already present on machine
22h         Normal    Created      pod/web-app-65967466dd-lxqcd        Created container: web-app
22h         Normal    Started      pod/web-app-65967466dd-lxqcd        Started container web-app
29m         Normal    SandboxChanged pod/web-app-65967466dd-lxqcd        Pod sandbox changed, it will be killed and re-created.
29m         Normal    Pulled       pod/web-app-65967466dd-lxqcd        Container image "python:3.11-slim" already present on machine
29m         Normal    Created      pod/web-app-65967466dd-lxqcd        Created container: web-app
```

#Información completa del deployment

Kubectl describe deployment web-app -n load-balancer-demo

```
alumno04@A6Alumno04:~$ kubectl describe deployment web-app -n load-balancer-demo
Name: web-app
Namespace: load-balancer-demo
CreationTimestamp: Thu, 22 Jan 2026 14:51:58 +0100
Labels: app=web-app
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=web-app
Replicas: 5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=web-app
  Containers:
    web-app:
      Image: python:3.11-slim
      Port: 5000/TCP
      Host Port: 0/TCP
      Command:
        sh
        -c
      Args:
        cd /app
        pip install --no-cache-dir -r requirements.txt > /dev/null 2>&1
        python app.py
  Liveness: http-get http://:5000/health delay=15s timeout=1s period=10s #success=1 #failure=3
  Readiness: http-get http://:5000/health delay=5s timeout=1s period=5s #success=1 #failure=3
  Environment:
    POD_NAME: (v1:metadata.name)
    POD_NAMESPACE: (v1:metadata.namespace)
  Mounts:
    /app from app-volume (rw)
  Volumes:
    app-volume:
      Type: ConfigMap (a volume populated by a ConfigMap)
      Name: app-files
      Optional: false
      Node-Selectors: <none>
      Tolerations: <none>
  Conditions:
    Type      Status  Reason
    ----      -
    Progressing  True    NewReplicaSetAvailable
    Available    True    MinimumReplicasAvailable
  OldReplicaSets: <none>
  NewReplicaSet: web-app-65967466dd (5/5 replicas created)
  Events:
    Type      Reason      Age   From      Message
    ----      -
    Normal    ScalingReplicaSet   21h   deployment-controller  Scaled up replica set web-app-65967466dd from 3 to 5
```

#Historial de cambios.

Kubectl rollout history deployment web-app -n load-balancer-demo.

```
alumno04@A6Alumno04:~$ kubectl rollout history deployment web-app -n load-balancer-demo
deployment.apps/web-app
REVISION  CHANGE-CAUSE
1          <none>
```

PARTE 8. ELIMINAR RECURSOS

8.1 - Limpiar Kubernetes.

#Eliminar todo el namespace.

Kubectrl delete namespace load-balancer-demo

#Verificar

Kubectrl get namespaces

NO TENGO CAPTURA, PERO EL FUNCIONAMIENTO FUE CORRECTO.

8.2 – Eliminar instancia EC2

En AWS Console:

1. **EC2** → Instances
2. Selecciona kubernetes-test-server
3. **Instance State** → Terminate
4. Confirma

Terminar (eliminar) instancia

⚠ En una instancia respaldada por EBS, la acción predeterminada se aplica al volumen de EBS raíz que se eliminará cuando se termine la instancia. El almacenamiento en las unidades locales se perderá.

¿Está seguro de que desea terminar estas instancias?

ID de la instancia	Protección de terminación
 i-056f56cf22cf3b7f6 (kubernetes-test-server)	 desactivado

Para confirmar que desea terminar las instancias, elija el botón de terminar que aparece a continuación. Las instancias que tengan la protección de terminación activada no se terminarán. La terminación de la instancia no se puede deshacer.

Omitir el apagado del sistema operativo
Esta opción omite el proceso de apagado controlado del sistema operativo. Utilícela solo cuando deba detener su instancia inmediatamente, como durante una emergencia o una conmutación por error.

☐ Omitir el apagado del sistema operativo

Cancelar Terminar (eliminar)

Instancias (1/3)

Información

Conectar

Estado de la instancia

Acciones

Lanzar instancias

Buscar Instancia por atributo o etiqueta (case-sensitive)

Todos los estados

<

1

>

	Name	ID de la instancia	Estado de la i...	Tipo de inst...	Comprob
	mi-servidor-web	i-04215907f6d9c8e93	En ejecución	t3.micro	Iniciali
	aws-wordpress	i-00f74f7f4aef547c3	En ejecución	t3.micro	Iniciali
	kubernetes-te...	i-056f56cf22cf3b7f6	Cerrándose	t3.micro	Iniciali

8.3 – Detener k3s

Si quieres pausar k3s

`sudo systemctl stop k3s`

```
alumno04@A6Alumno04:~$ sudo systemctl stop k3s
[sudo] password for alumno04:
alumno04@A6Alumno04:~$ _
```

Para eliminar completamente

`sudo /usr/local/bin/k3s-uninstall.sh`

```
alumno04@A6Alumno04:~$ sudo /usr/local/bin/k3s-uninstall.sh
+ id -u
+ [ 0 -eq 0 ]
+ K3S_DATA_DIR=/var/lib/rancher/k3s
+ /usr/local/bin/k3s-killall.sh
+ [ -s /etc/systemd/system/k3s.service ]
+ basename /etc/systemd/system/k3s.service
+ systemctl stop k3s.service
+ [ -x /etc/init.d/k3s* ]
+ killtree 1585 1724 1753 2370 2466
+ kill -9 1585 1884 2527 1724 1918 2680 1753 1907 2568 2370 2473 2807 2956 2466 2511 2843
+ do_unmount_and_remove /run/k3s
+ set +x
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/fe943e7f44843cdbf79fa6efcd2f7e9358c052f213e93002f00aaec49a76a5b/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/e65feeb82666af37cc2c2ecc9776c65dfa0205f50c31fbaa3a97cecb757aec51/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/bcc43f66995b4af704a5382b559dd020483c750e899a22610da32c74cdf217c4/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/a68c45db51092baf1859b84f25e8c5c793ee83b999fe3fca991e98c2247e7876/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/80625d8470488df2e4e041fc4cce6b847d4e2c7c516098c078becb31042ff841/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/769a659e12ae9a3132550df99134c54c93b9d1cfb6ff55a754322513dfb4e96b/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/6af2f9203f135ab62f3a250cc8c7b1f09d5daf021fb214f0babbcff451dee628/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/2ef8494fee0b876e175775bb1993b7673ecc879919a55692e0cd5205f65bd60b/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/24a40a76a6e351b931c9c3489b14ea1c4f9030af5e99713ef3cef7fbcba13a17/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/222d8000d5cf158af15f09a154e56f3b344553361b72bb30bdaab647bc600f8f/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.runtime.v2.task/k8s.io/05ff5b1260291afec3020df5b2dbd242e7321f4f3aff923c8136a7f7859d1b54/rootfs'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.grpc.v1.cri/sandboxes/fe943e7f44843cdbf79fa6efcd2f7e9358c052f213e93002f00aaec49a76a5b/shm'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.grpc.v1.cri/sandboxes/e65feeb82666af37cc2c2ecc9776c65dfa0205f50c31fbaa3a97cecb757aec51/shm'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.grpc.v1.cri/sandboxes/a68c45db51092baf1859b84f25e8c5c793ee83b999fe3fca991e98c2247e7876/shm'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.grpc.v1.cri/sandboxes/769a659e12ae9a3132550df99134c54c93b9d1cfb6ff55a754322513dfb4e96b/shm'
sh -c 'umount -f "$0" && rm -rf "$0" /run/k3s/containerd/io.containerd.grpc.v1.cri/sandboxes/24a40a76a6e351b931c9c3489b14ea1c4f9030af5e99713ef3cef7fbcba13a17/shm'
+ do_unmount_and_remove /var/lib/kubelet/pods
+ set +x
```